

Disproof of Conjecture 00000000945

Real-valued Lipschitz functions on the circle extend

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The conjecture asserts that there exists a Lipschitz function $f : S^1 \rightarrow \mathbb{R}$ that extends to no Banach plane but extends to a three-dimensional target. Its obstruction already fails: every such function extends, preserving its Lipschitz constant, to the Euclidean plane.

Here $S^1 = \{x \in \mathbb{R}^2 : \|x\|_2 = 1\}$ has the metric induced from the Euclidean norm. An extension to a plane means a function defined on that plane agreeing with f on the embedded circle. We also discuss the alternative reading of “target” below.

Theorem 1. *Let (X, d) be a metric space, $A \subseteq X$ be nonempty, and let $f : A \rightarrow \mathbb{R}$ satisfy $|f(a) - f(b)| \leq Kd(a, b)$ for all $a, b \in A$, where $K \geq 0$. There is a function $F : X \rightarrow \mathbb{R}$ agreeing with f on A and satisfying the same Lipschitz bound.*

Proof. Define

$$F(x) = \inf_{a \in A} (f(a) + Kd(x, a)).$$

For a fixed $a_0 \in A$, the triangle inequality and the Lipschitz bound give

$$f(a) + Kd(x, a) \geq f(a_0) - Kd(a, a_0) + Kd(x, a) \geq f(a_0) - Kd(x, a_0).$$

The family defining the infimum is nonempty and bounded below; taking $a = a_0$ bounds it above as well. Thus $F(x)$ is a finite real number. For $x \in A$, every term is at least $f(x)$, and the term $a = x$ equals $f(x)$, so $F(x) = f(x)$.

For every $a \in A$, $d(x, a) \leq d(y, a) + d(x, y)$. Taking infima yields $F(x) \leq F(y) + Kd(x, y)$. Exchanging x, y proves $|F(x) - F(y)| \leq Kd(x, y)$. $\square$

Apply the theorem with $X = \mathbb{R}^2$, $A = S^1$. The Euclidean plane is a two-dimensional real Banach space. Consequently no Lipschitz $f : S^1 \rightarrow \mathbb{R}$ has the claimed obstruction. This negates a necessary conjunct of the existential statement, irrespective of its additional three-dimensional extension requirement.

If “Banach plane” instead describes the codomain, embed $\mathbb{R}$ isometrically as $t \mapsto (t, 0)$ in the Banach plane $(\mathbb{R}^2, \|\cdot\|_\infty)$. The map $x \mapsto (F(x), 0)$ agrees with the embedded original function and is K -Lipschitz. Thus this reading also supplies a plane-valued extension. Without specifying an embedding or superspace, a different meaning of extension would require additional data absent from the statement.

Formal verification. The accompanying Lean project uses actual functions on the subtype of the Euclidean unit sphere, not scalar substitutes or assumed extension certificates. `subset_extension` proves the general metric-space result using Mathlib’s `LipschitzOnWith.extend_real`; `circle_extension` specializes it to the circle. `conjecture_false` negates the existential failure of any Lipschitz extension to the plane, and `plane_target_extension` verifies the codomain interpretation. The final theorems’ axiom audits use only standard Lean axioms. The Mathlib implementation contains the infimum proof above.